

Final Project Report: How Music Taste Changes Over Time

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Abstract

Music is a key part of daily life, yet people's opinions about music have evolved over time. With factors, such as race, having a different impact on society during each decade, this study focuses on how various factors affect song ratings during different decades. Data from Billboard Hot 100 Number One songs were taken from the 1960s - 2010s and split up to form rating prediction linear models. We compared main effects models with interaction effects models and decided which one would be more viable to optimize. The chosen model for the decade was optimized based on its AIC score to determine which predictors made a significant impact on the model. Comparing the predictors across models revealed that variables such as energy and race were consistently important across most decades. However, no singular factor was consistent between every decade. Ultimately, the findings demonstrated that while the factors that predict song ratings change over time, they have the potential to be similar in different time periods.

1 Introduction

Introduce your research question and the motivation behind your study. Why is this topic interesting or important? State the primary objectives of your regression analysis.

We have 3 main research questions. They include:

1. How do demographic characteristics of the artist affect the overall rating of a song over time?
2. How do song characteristics (provided by Spotify) affect the overall rating of a song over time?

3. Which demographics and song characteristics significantly impact the overall rating of a Billboard #1 song over the 6 decades?

We were interested in how music has changed over time and we thought separating our data into different time periods helps us understand what characteristics truly impact the overall rating. We thought each decade had its own music taste, so we decided to analyze the data based on each decade the hits were created. It is interesting to see how far music has gone over the years, as well as getting a better grasp of music taste in each decade.

2 Data Description and Exploratory Data Analysis (EDA)

Describe the dataset you are using. Where did it come from? What are the observational units?

2.1 Variables

Briefly explain the response variable (Y) and the main predictor variables (X_i).

Let Y denote the overall rating of the song. The primary predictors are:

- X_1 : Age: Average age of the artist(s)
- X_2 : Gender: Gender of the artist (s)
- X_3 : Race: Race of the artist (s)
- X_4 : Genre: Type of genre of the song
- X_5 : Energy: Energy measure from Spotify (0-100)
- X_6 : Danceability: Danceability measure from Spotify (0-100)
- X_7 : Happiness: Happiness measure from Spotify (0-100)
- X_8 : Loudness_{db} : *Loudness measured in decibels from Spotify*
- X_9 : BPM: Beats per minute from Spotify
- X_{10} : Length_{sec} : *Length of the song in seconds*

2.2 Exploratory Analysis

Present initial findings from your EDA. You can reference figures like scatterplots or correlation matrices here (e.g., "As seen in Figure 1...").

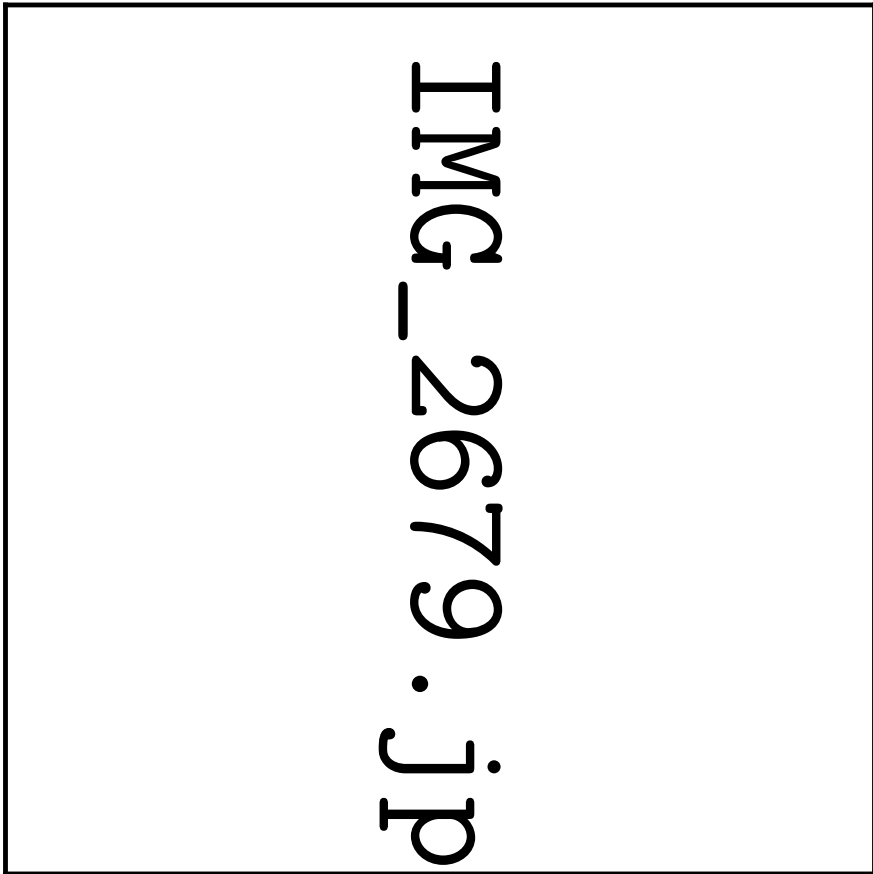


Figure 1: A descriptive caption explaining the plot.

3 Methodology and Model Building

Describe the models you fit to the data.

3.1 Proposed Model

We fit a multiple linear regression model of the form:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \epsilon_i \quad (1)$$

where $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Explain how you selected your variables (e.g., stepwise selection, theoretical justification, handling of multicollinearity). If you used transformations (like taking the log of the response), explain why.

4 Results and Diagnostics

Present the results of your final model.

4.1 Model Summary

Table 1 summarizes the estimated coefficients for our final model.

Table 1: Estimated Regression Coefficients and Significance

Predictor	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	12.45	2.10	5.93	< 0.001
X_1	3.12	0.45	6.93	< 0.001
X_2	-1.05	0.30	-3.50	0.002

4.2 Model Diagnostics

Assess the assumptions of linear regression (linearity, independence, homoscedasticity, normality of residuals). You should include residual plots or QQ-plots here to justify that your model assumptions hold (or discuss how they are violated).

5 Discussion and Conclusion

Summarize the main takeaways from your model in the context of the original problem. What do the coefficients mean in the real world?

Acknowledge the limitations of your study (e.g., potential omitted variable bias, non-representative sample) and suggest areas for future work.

References

References

- [1] Author Name. *Title of the Textbook*. Publisher, Year.
- [2] Dataset Creator. “Title of the Dataset.” *Source/URL*, Year.

A Appendix: R/Python Code

Paste your analysis code here. If you prefer, you can use the `verbatim` environment to format it as raw text:

```
# Example R Code
model <- lm(Y ~ X1 + X2, data = mydata)
summary(model)
plot(model)
```